Heteroskedasticity

EC421, Set 04

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Prologue

R showcase

`Quarto` - Simple mark-up language for for combining/creating documents, equations, figures, R, and more - `Basics of Markdown` - *E.g.*, `**I'm bold**`, `*I'm italic*`, `I = "code"`

Interactive data visualization beyond `shiny`: `plotly`, `crosstalk`, `echarts4r`, `highcharter`.

Econometrics with R - (Currently) free, online textbook - Written and published using R (and probably `R Markdown`) - *Warning:* I haven't read the full book.

Related: Tyler Ransom has a great cheatsheet for econometrics.

Schedule

Last Time

We wrapped up our review.

Today

Heteroskedasticity

Schedule

This week

First assignment!

Submit **one file** (HTML, PDF, or Word) that includes

1. Your written answers
2. Figures and regression output
3. The R code that generated your answers.

This file should be a rendered RMarkdown or Quarto file (`.rmd` or `.qmd`).

Important

- We should be able to easily find your answers for each question.
- **Do not copy.** (You **will** receive a zero.)

Prerequisite material

Prerequisite material

The χ^2 distribution

Some test statistics are distributed as χ^2 random variables.

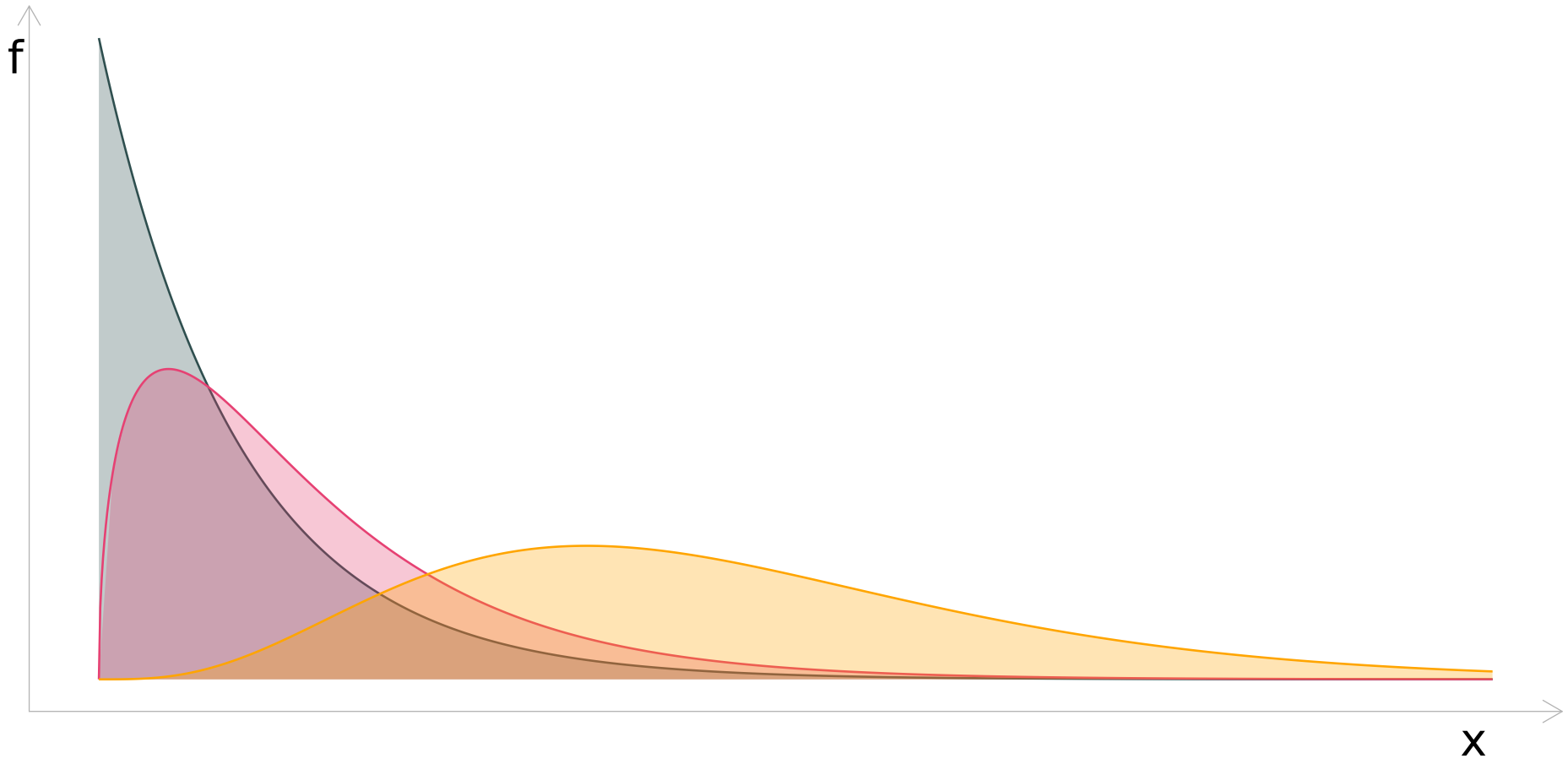
The χ^2 distribution is just another example of a common (named) distribution (like the Normal distribution, the t distribution, and the F).

The shape of the χ^2 distribution depends on a single parameter: - We will call this parameter k - Our test statistics will refer to k as *degrees of freedom*.

Prerequisite material

The χ^2 distribution

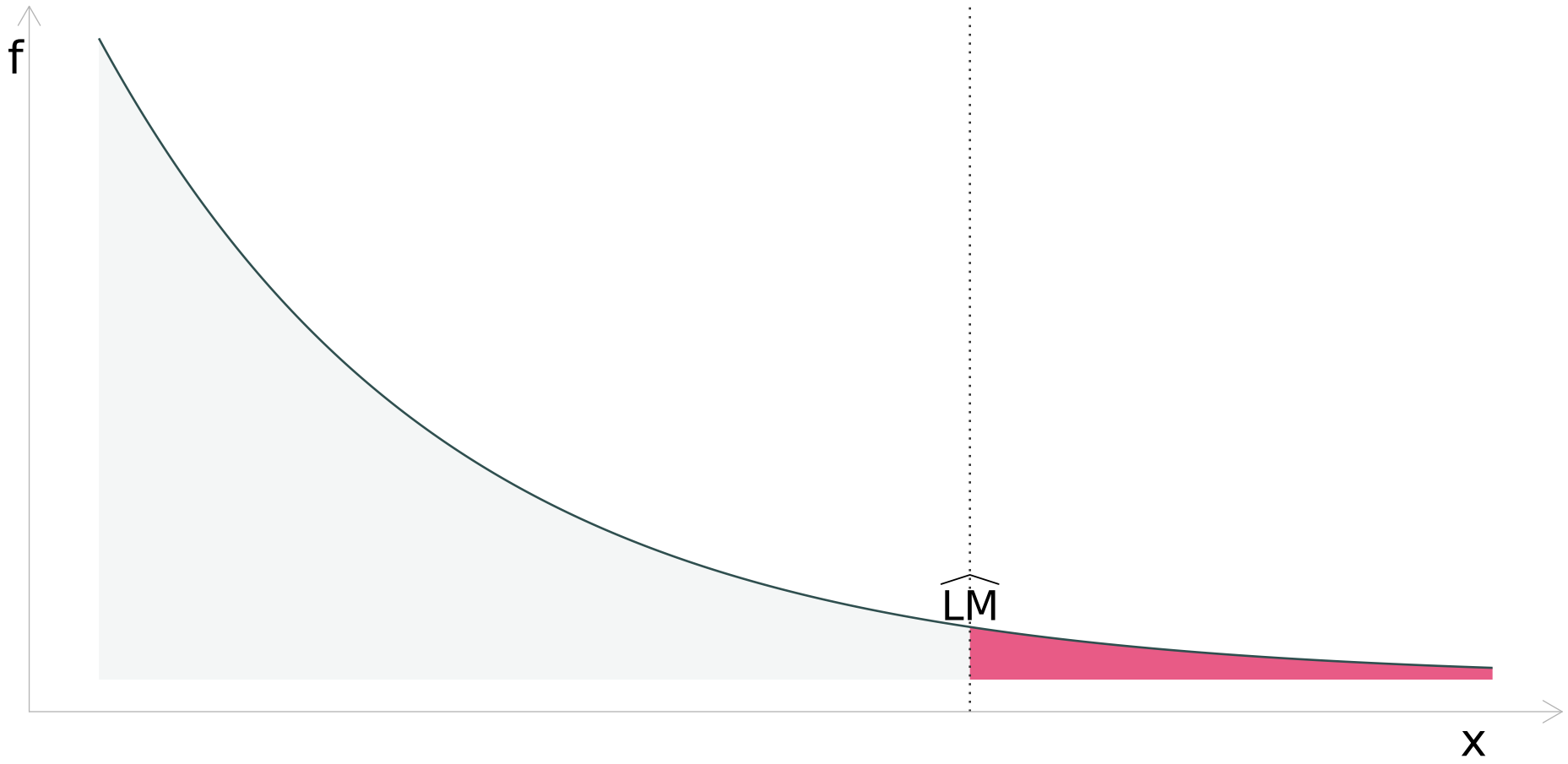
Three examples of χ^2_k: $k = 1$, $k = 2$, and $k = 9$



Prerequisite material

The χ^2 distribution

Probability of observing a more extreme test statistic $\widehat{\mathbf{LM}}$ under H_0



Heteroskedasticity

Heteroskedasticity

Let's write down our **current assumptions**

Heteroskedasticity

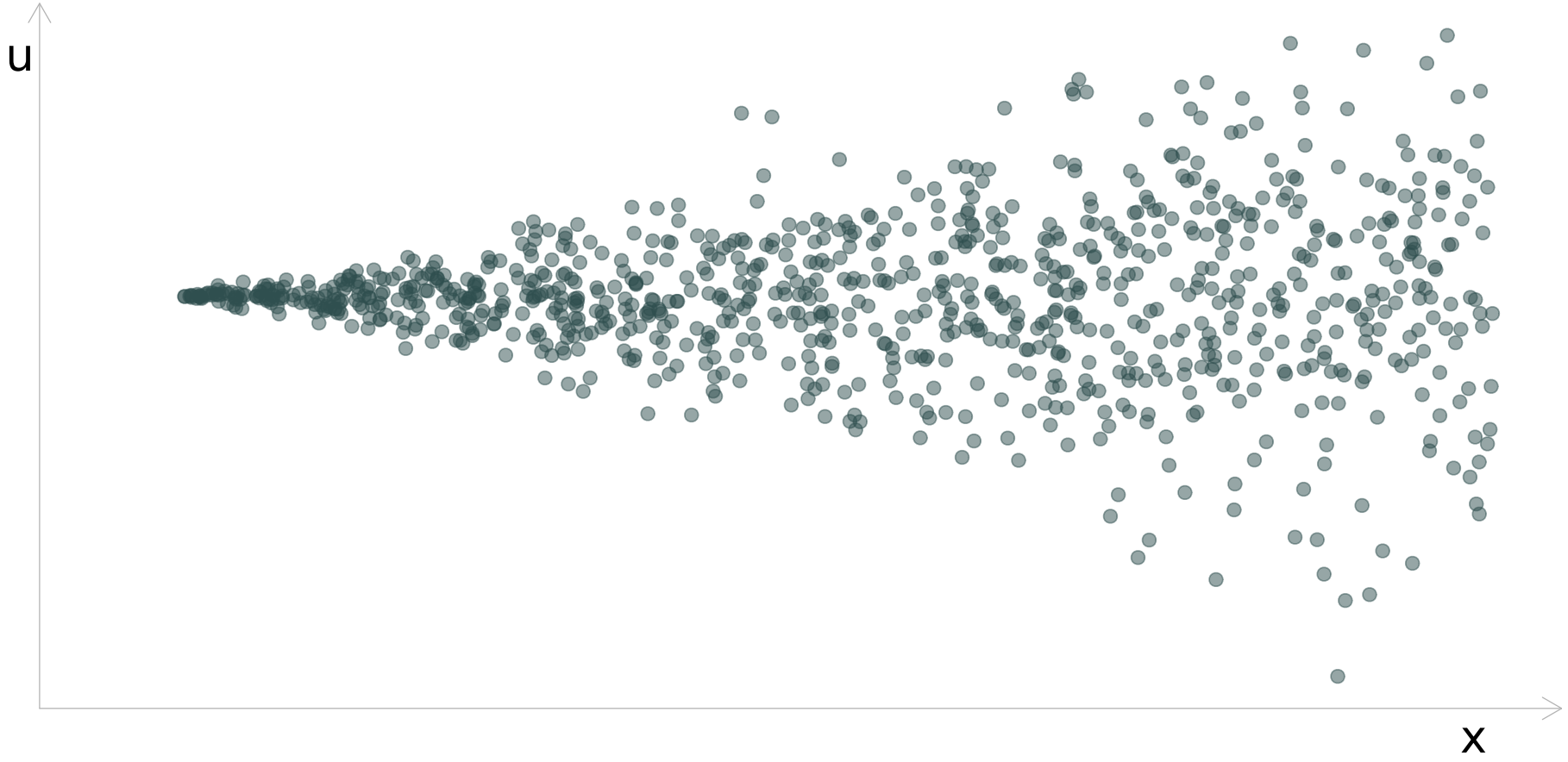
Today we're focusing on assumption #5:

> 5. The disurbances have **constant variance** σ^2 and **zero covariance**, *i.e.*, -
> $\boldsymbol{E}\left[u_i^2|X\right] = \text{Var}\left(u_i|X\right) = \sigma^2 \implies \text{Var}\left(u_i\right) = \sigma^2$ -
> $\text{Cov}\left(u_i,\, u_j|X\right) = \boldsymbol{E}\left[u_i u_j|X\right] = 0$ for $i \neq j$

Heteroskedasticity

Classic example of heteroskedasticity: The funnel

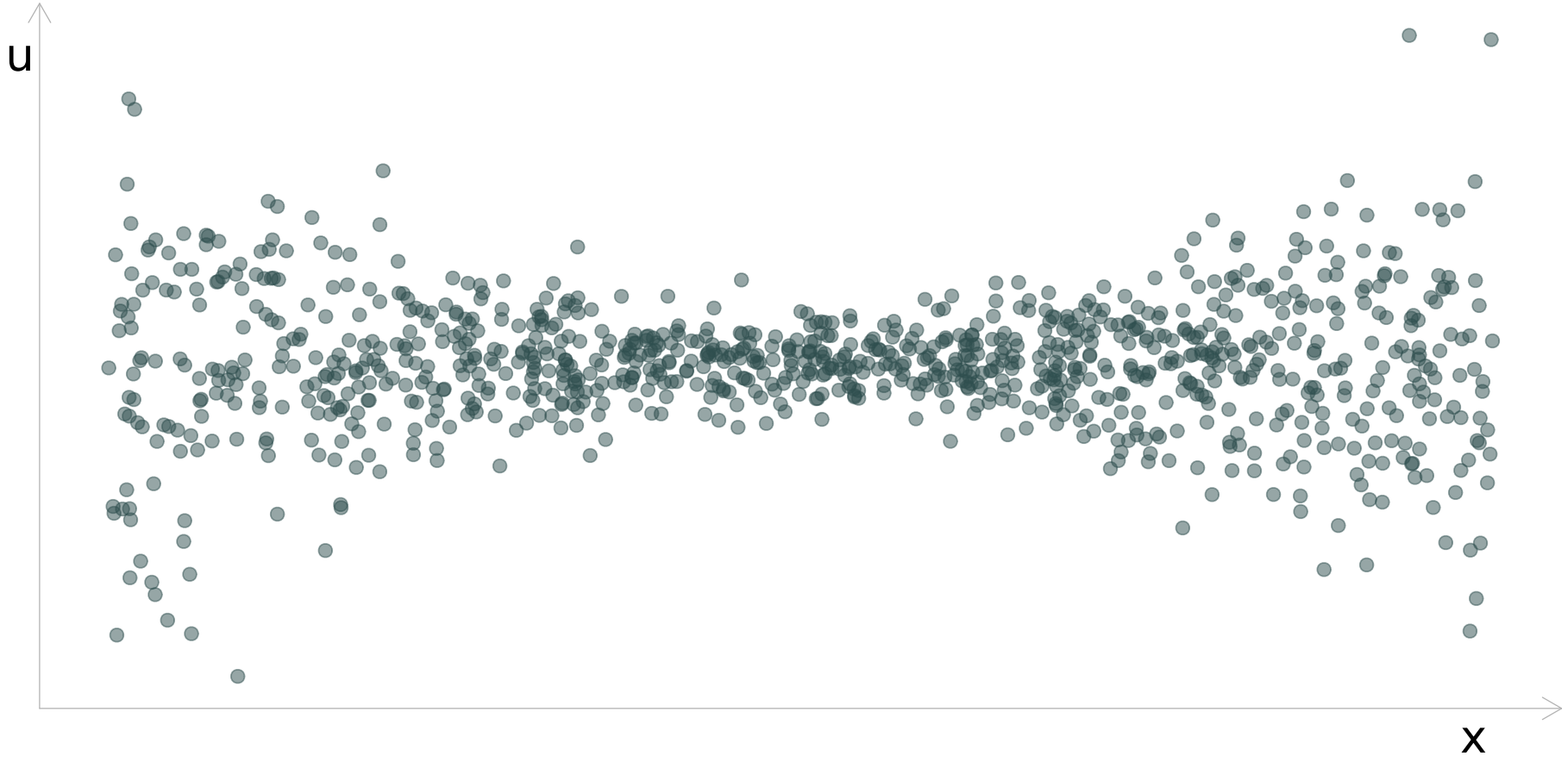
Variance of u increases with x



Heteroskedasticity

Another example of heteroskedasticity: (double funnel?)

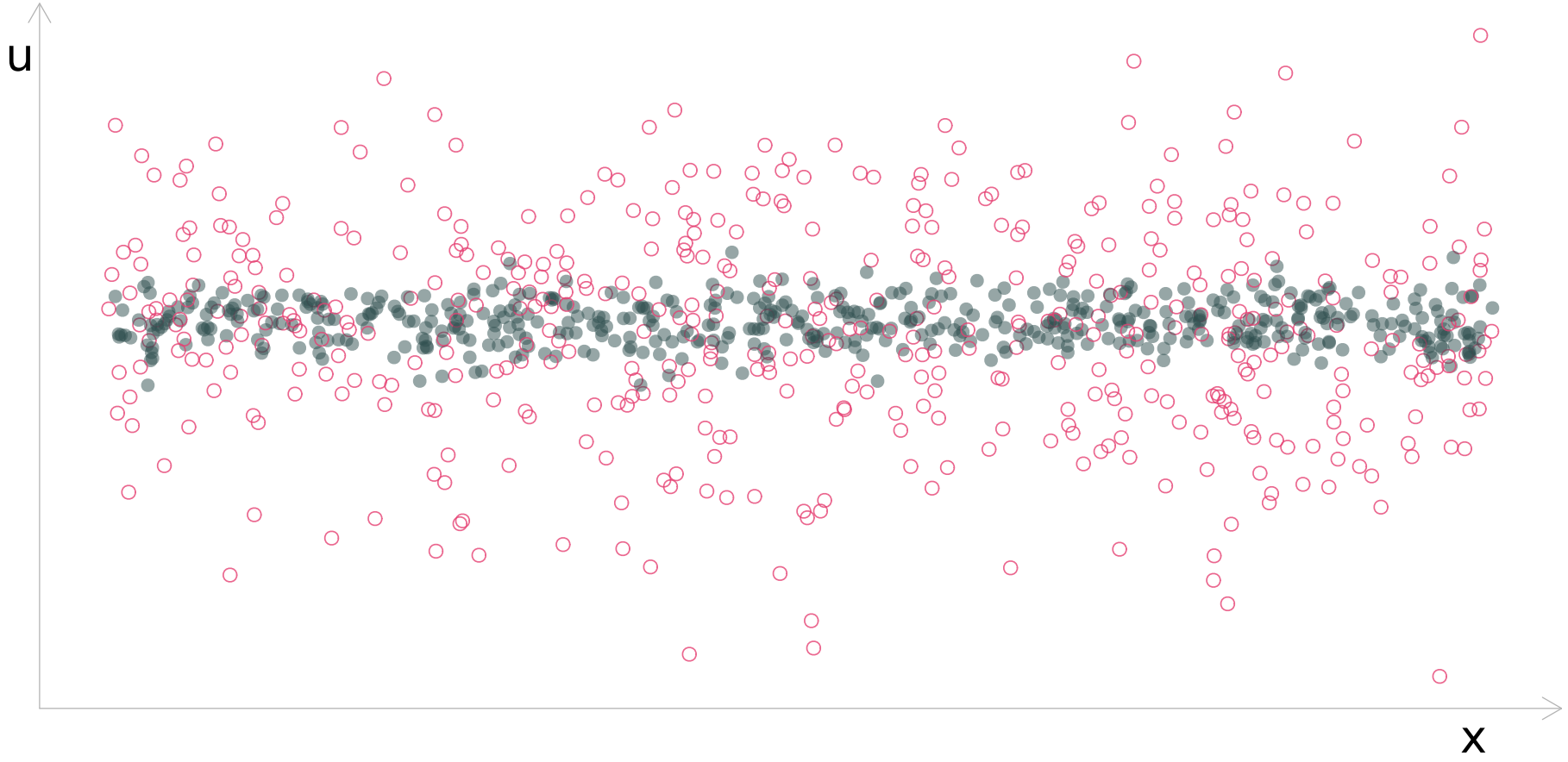
Variance of u increasing at the extremes of x



Heteroskedasticity

Another example of heteroskedasticity:

Differing variances of u by group



Heteroskedasticity

What and why?

Heteroskedasticity is present when the variance of u changes with any combination of our explanatory variables x_1, through x_k (henceforth: X).

Heteroskedasticity

Consequences

So what are the consquences of heteroskedasticity? Bias? Inefficiency?

First, let's check if it has consquences for the the unbiasedness of OLS.

Heteroskedasticity

Proof: Assuming $y_i = \beta_0 + \beta_1 x_i + u_i$

$$
\begin{aligned}
\hat{\beta}_1 &= \frac{\sum_i (y_i - \overline{y})(x_i - \overline{x})}{\sum_i (x_i - \overline{x})^2} \\
&= \frac{\sum_i \left([\beta_0 + \beta_1 x_i + u_i] - [\beta_0 + \beta_1 \overline{x} + \overline{u}]\right)(x_i - \overline{x})}{\sum_i (x_i - \overline{x})^2} \\
&= \frac{\sum_i \left(\beta_1 [x_i - \overline{x}] + [u_i - \overline{u}]\right)(x_i - \overline{x})}{\sum_i (x_i - \overline{x})^2} \\
&= \frac{\sum_i \left(\beta_1 [x_i - \overline{x}]^2 + [x_i - \overline{x}][u_i - \overline{u}]\right)}{\sum_i (x_i - \overline{x})^2} \\
&= \beta_1 + \frac{\sum_i (x_i - \overline{x})(u_i - \overline{u})}{\sum_i (x_i - \overline{x})^2}
\end{aligned}
$$

Heteroskedasticity

$$
\begin{aligned}
\hat{\beta}_1 = \cdots = \beta_1 + \frac{\sum_i (x_i - \overline{x})(u_i - \overline{u})}{\sum_i (x_i - \overline{x})^2} \\
= \beta_1 + \frac{\sum_i (x_i - \overline{x}) u_i - \overline{u} \sum_i (x_i - \overline{x})}{\sum_i (x_i - \overline{x})^2} \\
= \beta_1 + \frac{\sum_i (x_i - \overline{x}) u_i - \overline{u} \left(\sum_i x_i - \sum_i \overline{x}\right)}{\sum_i (x_i - \overline{x})^2} \\
= \beta_1 + \frac{\sum_i (x_i - \overline{x}) u_i - \overline{u} \left(\sum_i x_i - n\overline{x}\right)}{\sum_i (x_i - \overline{x})^2} \\
= \beta_1 + \frac{\sum_i (x_i - \overline{x}) u_i - \overline{u} \left(\sum_i x_i - \sum_i x_i\right)}{\sum_i (x_i - \overline{x})^2} \\
= \beta_1 + \frac{\sum_i (x_i - \overline{x}) u_i}{\sum_i (x_i - \overline{x})^2} \quad 😅
\end{aligned}
$$

Heteroskedasticity

Consequences: Bias

We now want to see if heteroskedasticity biases the OLS estimator for β_1.

Heteroskedasticity

Consequences: Efficiency

OLS's **efficiency** and **inference** do not survive heteroskedasticity.

- In the presence of heteroskedasticity, OLS is **no longer the most efficient** (best) linear unbiased estimator.

Heteroskedasticity

Consequences: Inference

OLS **standard errors are biased** in the presence of heteroskedasticity.

- Wrong confidence intervals
- Problems for hypothesis testing (both t and F tests)

Heteroskedasticity

Solutions

1. **Tests** to determine whether heteroskedasticity is present.
2. **Remedies** for (1) efficiency and (2) inference

Testing for heteroskedasticity

Testing for heteroskedasticity

While we *might* have solutions for heteroskedasticity, the efficiency of our estimators depends upon whether or not heteroskedasticity is present.

1. The **Goldfeld-Quandt test**
2. The **White test**

1. There are many other options for testing, *e.g.*, the **Breusch-Pagan test**.

Testing for heteroskedasticity

The Goldfeld-Quandt test

Focuses on a specific type of heteroskedasticity: whether the variance of u_i differs **between two groups**.[1]

Remember how we used our residuals to estimate the σ^2?

$$s^2 = \frac{\text{SSE}}{n-1} = \frac{\sum_i e_i^2}{n-1}$$

We will use this same idea to determine whether there is evidence that our two groups differ in the variances of their disturbances, effectively comparing s_1^2 and s_2^2 from our two groups.

1. The G-Q test was one of the early tests of heteroskedasticity (1965).

Testing for heteroskedasticity

The Goldfeld-Quandt test

Operationally,

1. Order your the observations by x
2. Split the data into two groups of size $n^\star$
   - G_1: The first third
   - G_2: The last third
3. Run separate regressions of y on x for G_1 and G_2
4. Record SSE_1 and SSE_2
5. Calculate the G-Q test statistic

Testing for heteroskedasticity

The Goldfeld-Quandt test

The G-Q test statistic

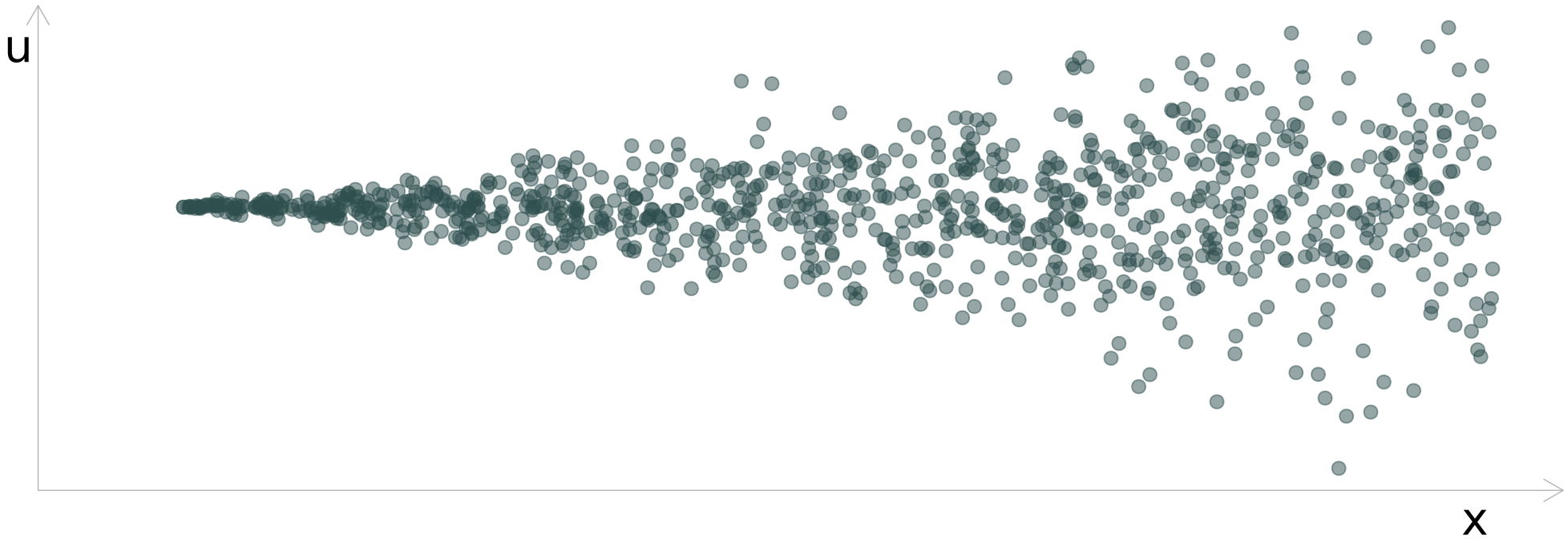
$$F_{(n^\star - k,\, n^\star - k)} = \frac{\text{SSE}_2/(n^\star - k)}{\text{SSE}_1/(n^\star - k)} = \frac{\text{SSE}_2}{\text{SSE}_1}$$

follows an F distribution (under the null hypothesis) with $n^\star - k$ and $n^\star - k$ degrees of freedom.[1]

1. Goldfeld and Quandt suggested $n^\star$ of $(3/8)n$. k gives number of estimated parameters (*i.e.*, $\hat{\beta}_j$'s).

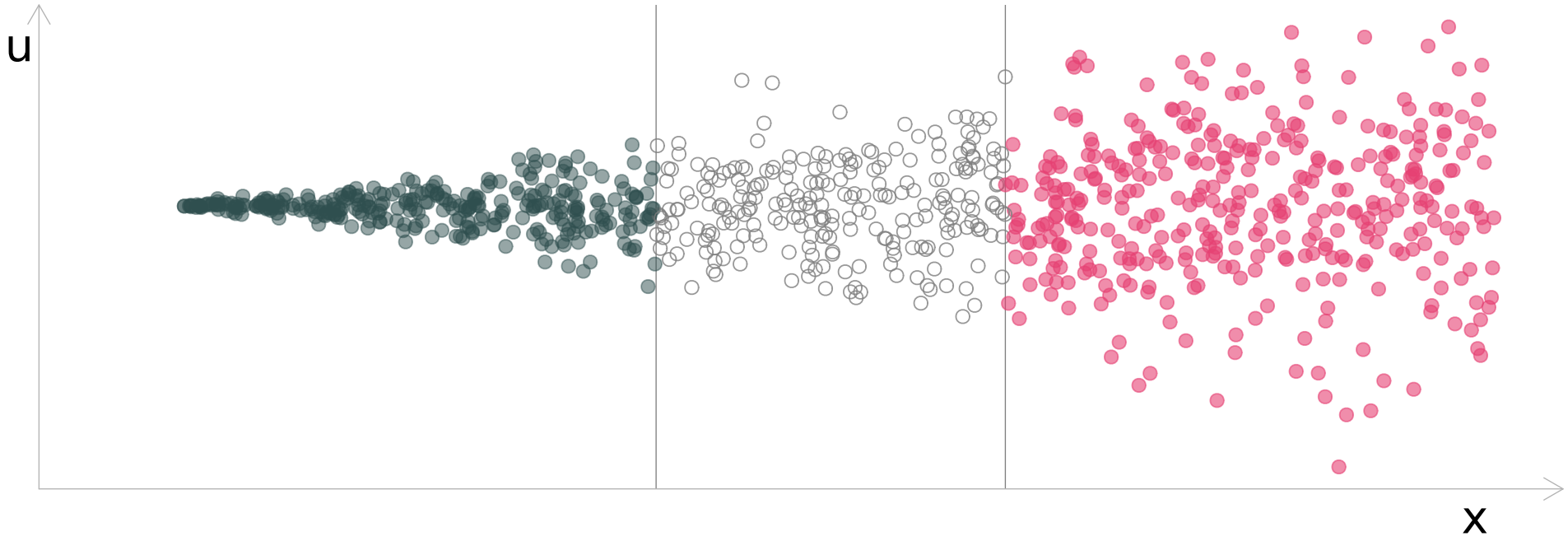
Testing for heteroskedasticity

The Goldfeld-Quandt test



Testing for heteroskedasticity

The Goldfeld-Quandt test



$$F_{375,\,375} = \frac{\mathrm{SSE}_2 = 18,203.4}{\mathrm{SSE}_1 = 1,039.5} \approx 17.5 \implies p\text{-value} < 0.001$$

$\therefore$ We reject H_0: $\sigma_1^2 = \sigma_2^2$ and conclude there is statistically significant evidence of heteroskedasticity.

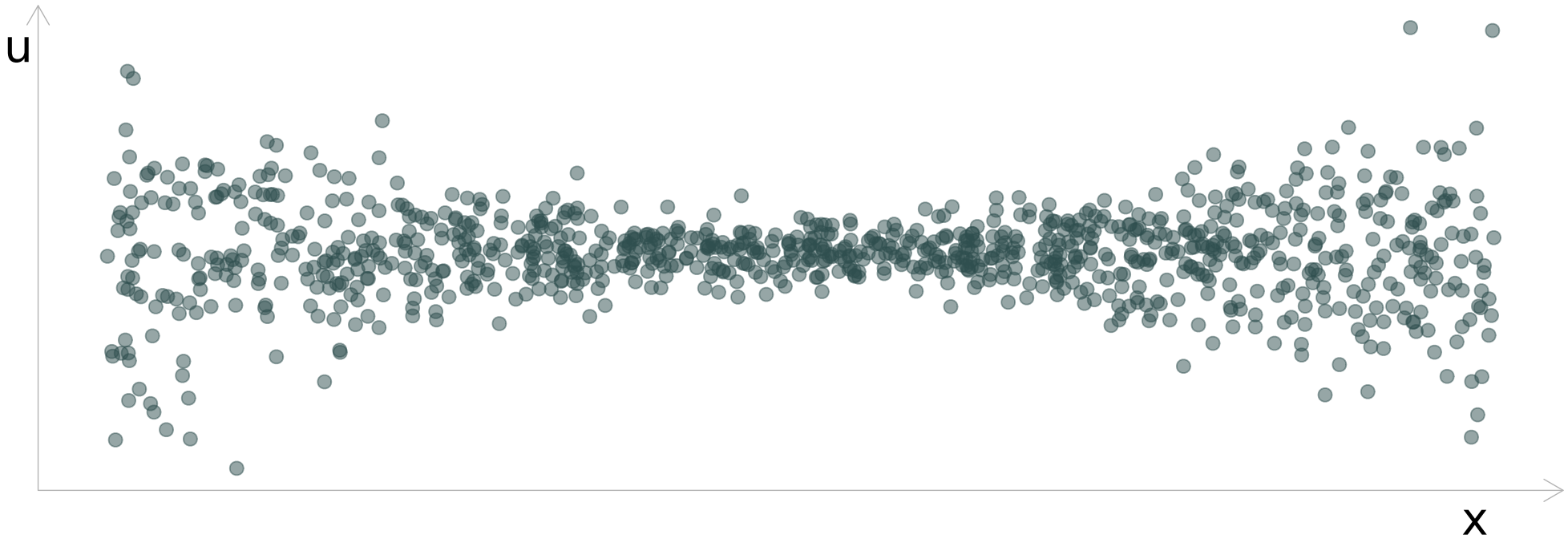
Testing for heteroskedasticity

The Goldfeld-Quandt test

The problem...

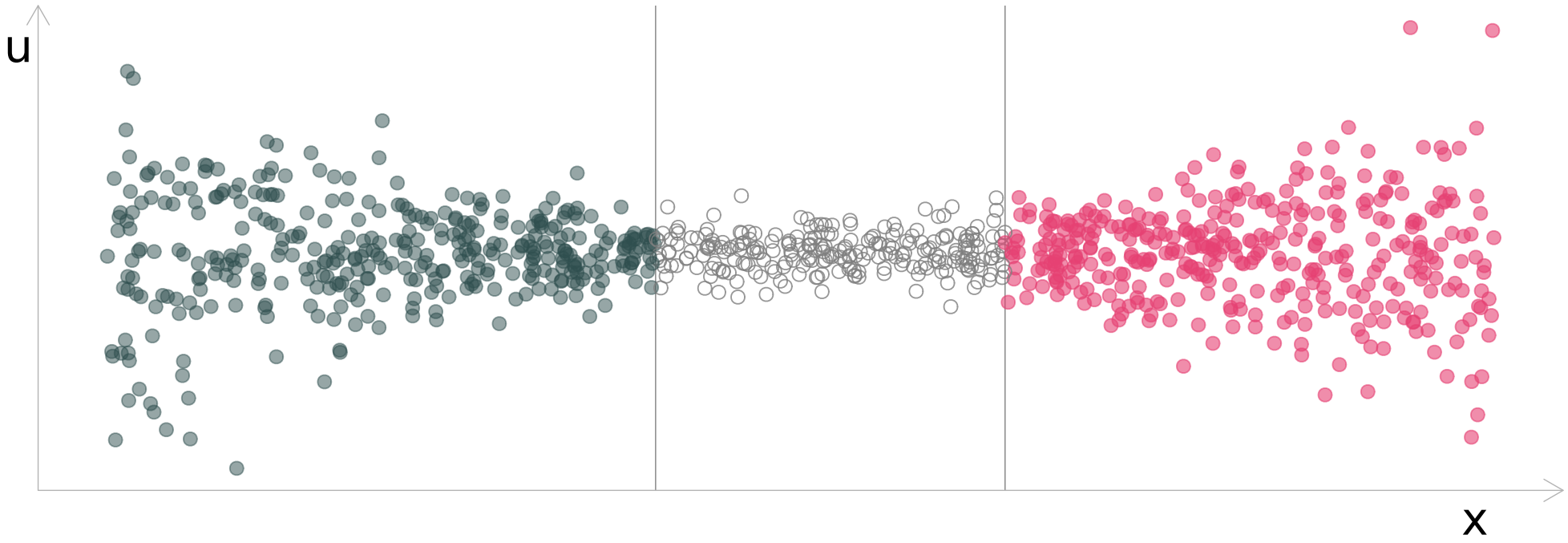
Testing for heteroskedasticity

The Goldfeld-Quandt test



Testing for heteroskedasticity

The Goldfeld-Quandt test



$$F_{375,\,375} = \frac{\mathrm{SSE}_2 = 14,516.8}{\mathrm{SSE}_1 = 14,937.1} \approx 1 \implies p\text{-value} \approx 0.609$$

$\therefore$ We fail to reject H_0: $\sigma_1^2 = \sigma_2^2$ while heteroskedasticity is present.

Testing for heteroskedasticity

The White test

Breusch and Pagan (1981) attempted to solve this issue of being too specific with the functional form of the heteroskedasticity.

- Regress e_i^2 on $X = [1, x_1, x_2, \ldots, x_k]$ and test for joint significance.
- Allows the data to show if/how the variance of u_i correlates with X.
- If σ_i^2 correlates with X, then we have heteroskedasticity.

Testing for heteroskedasticity

The White test

Toward this goal, Hal White took advantage of the fact that we can **replace the homoskedasticity requirement with a weaker assumption**:

- **Old:** $\text{Var}\left(u_i|X\right) = \sigma^2$
- **New:** u^2 is *uncorrelated* with the explanatory variables (*i.e.*, x_j for all j), their squares (*i.e.*, x_j^2), and the first-degree interactions (*i.e.*, $x_j x_h$).

Testing for heteroskedasticity

The White test

An outline of White's test for heteroskedasticity:

1. Regress y on x_1, x_2, ..., x_k. Save residuals e.
2. Regress squared residuals on all explanatory variables, their squares, and interactions.

$$e^2 = \alpha_0 + \sum_{h=1}^{k} \alpha_h x_h + \sum_{j=1}^{k} \alpha_{k+j} x_j^2 + \sum_{\ell=1}^{k-1} \sum_{m=\ell+1}^{k} \alpha_{\ell,m} x_\ell x_m + v_i$$

3. Record R_e^2.
4. Calculate test statistic to test H_0: $\alpha_p = 0$ for all $p \neq 0$.

Testing for heteroskedasticity

The White test

White's test statistic is

$$\text{LM} = n \times R_e^2 \qquad \text{Under H}_0\text{, LM} \overset{\text{d}}{\sim} \chi_k^2$$

where R_e^2 comes from the regression of e^2 on the explanatory variables, their squares, and their interactions.

$$e^2 = \alpha_0 + \underbrace{\sum_{h=1}^{k} \alpha_h x_h}_{\text{Expl. variables}} + \underbrace{\sum_{j=1}^{k} \alpha_{k+j} x_j^2}_{\text{Squared terms}} + \underbrace{\sum_{\ell=1}^{k-1} \sum_{m=\ell+1}^{k} \alpha_{\ell,m} x_\ell x_m}_{\text{Interactions}} + v_i$$

Note: The k (for our χ_k^2) equals the number of estimated parameters in the regression above (the α_j), excluding the intercept (α_0).

Testing for heteroskedasticity

The White test

Practical note: If a variable is equal to its square (*e.g.*, binary variables), then you don't (can't) include it. The same rule applies for interactions.

Testing for heteroskedasticity

The White test

Example: Consider the model[1] $y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 + u$

Step 1: Estimate the model; obtain residuals (e).

Step 2: Regress e^2 on explanatory variables, squares, and interactions.

$$\begin{aligned} e^2 =& \alpha_0 + \alpha_1 x_1 + \alpha_2 x_2 + \alpha_3 x_3 + \alpha_4 x_1^2 + \alpha_5 x_2^2 + \alpha_6 x_3^2 \\ &+ \alpha_7 x_1 x_2 + \alpha_8 x_1 x_3 + \alpha_9 x_2 x_3 + v \end{aligned}$$

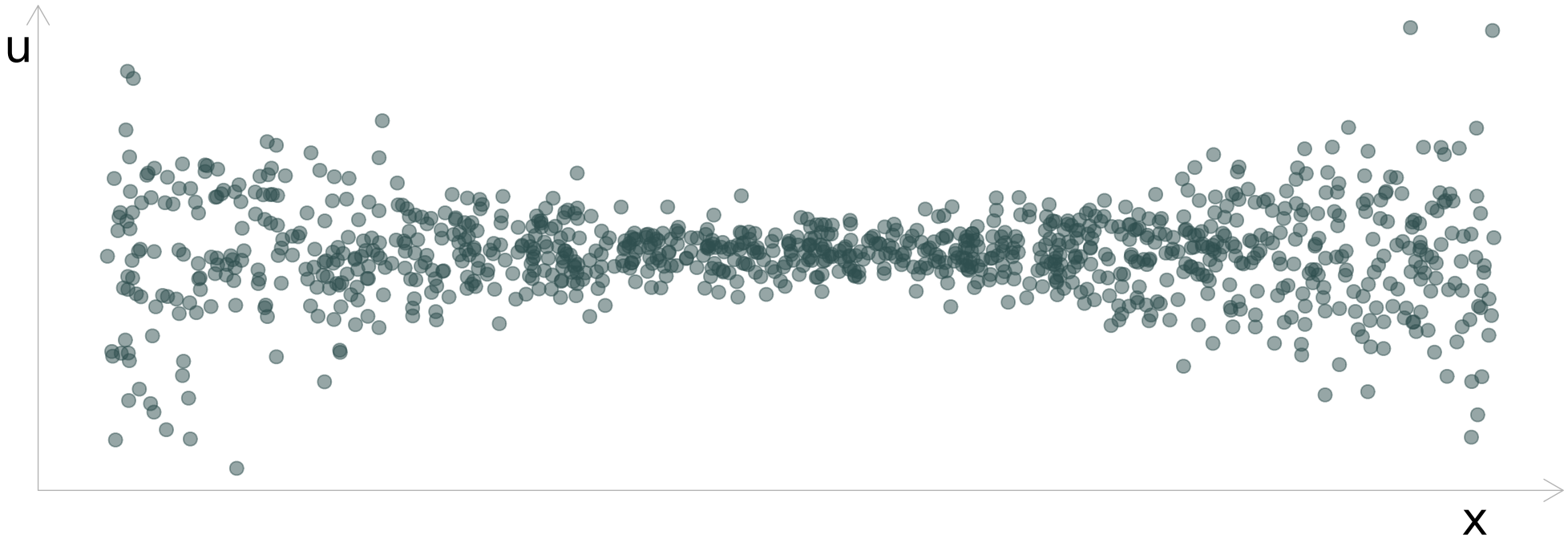
Record the R^2 from this equation (call it R_e^2).

Step 3: Test H_0: $\alpha_1 = \alpha_2 = \cdots = \alpha_9 = 0$ using $\text{LM} = nR_e^2 \overset{\text{d}}{\sim} \chi_9^2$.

1. To simplify notation here, I'm dropping the i subscripts.

Testing for heteroskedasticity

The White test



The White test for this simple linear regression.

$$e_i^2 = \hat{\alpha}_0 + \hat{\alpha}_1 x_{1i} + \hat{\alpha}_2 x_{1i}^2 \qquad \widehat{\text{LM}} = 185.8 \qquad p\text{-value} < 0.001$$

Testing for Heteroskedasticity

Examples

Testing for heteroskedasticity

Examples

Goal: Estimate the relationship between standardized test scores (outcome variable) and (1) student-teacher ratio and (2) income, *i.e.*,

$$(\text{Test score})_i = \beta_0 + \beta_1 \text{Ratio}_i + \beta_2 \text{Income}_i + u_i \tag{1}$$

Potential issue: Heteroskedasticity... and we do not observe u_i.

Solution:

1. Estimate the relationship in (1) using OLS.
2. Test for heteroskedasticity.

- Goldfeld-Quandt
- White

Testing for heteroskedasticity

Examples

We will use testing data from the dataset `Caschool` in the `Ecdat` R package.

```
#> # A tibble: 2 × 3
#>   test_score ratio income
#>        <dbl> <dbl>  <dbl>
#> 1       691.  17.9  22.7
#> 2       661.  21.5   9.82
```

Testing for heteroskedasticity

Examples

Let's begin by estimating our model

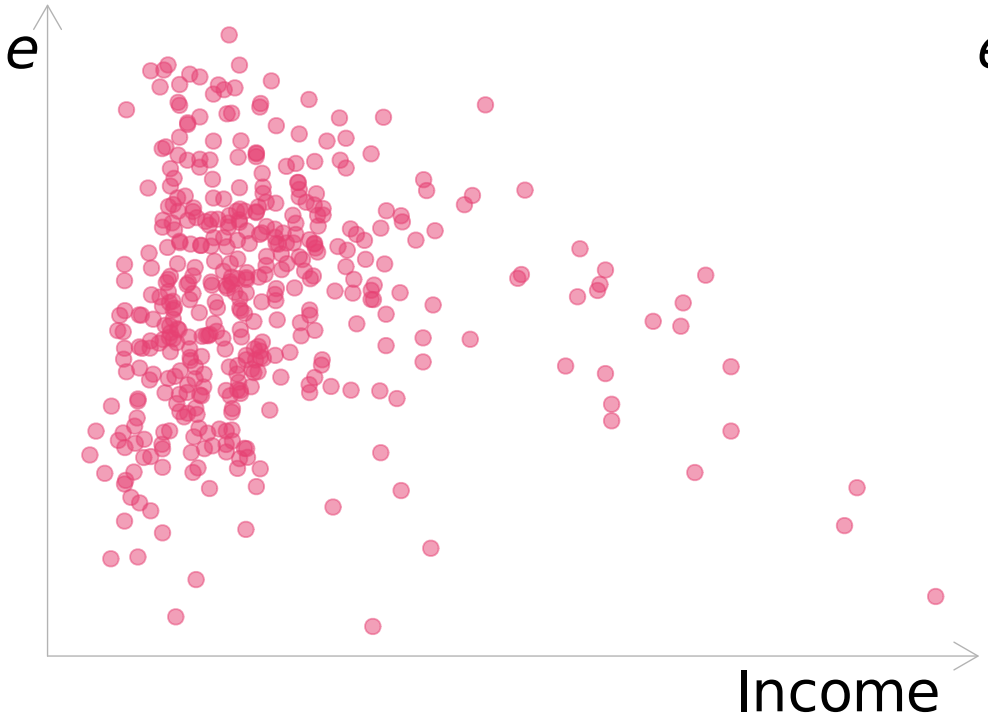
$$(\text{Test score})_i = \beta_0 + \beta_1 \text{Ratio}_i + \beta_2 \text{Income}_i + u_i$$

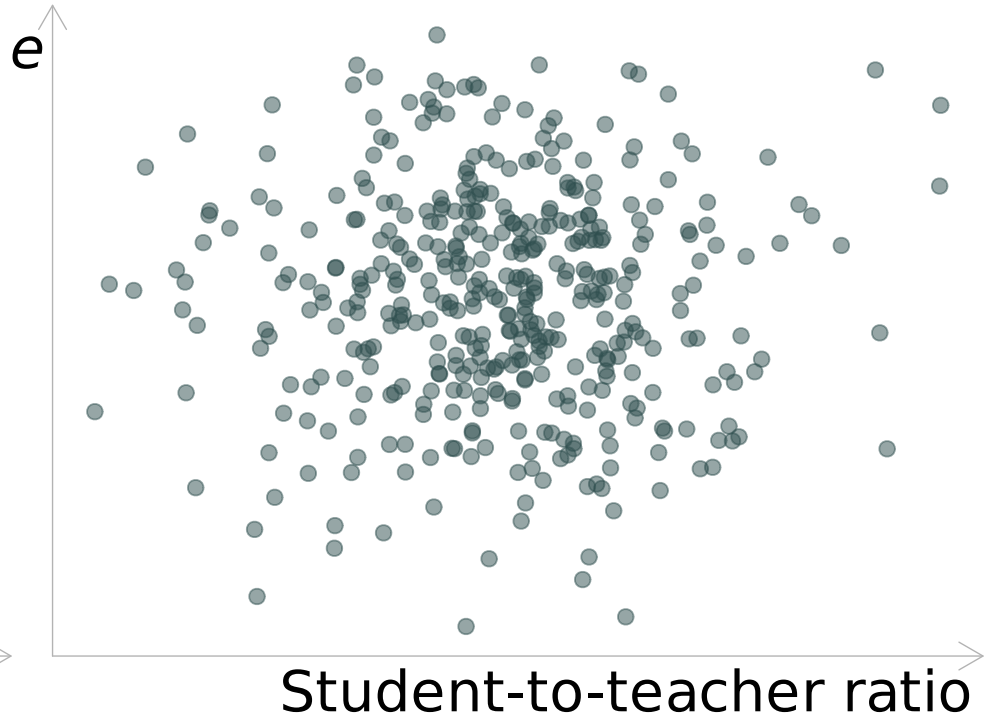
```
#> # A tibble: 3 × 5
#>   term        estimate std.error statistic   p.value
#>   <chr>          <dbl>     <dbl>     <dbl>     <dbl>
#> 1 (Intercept)  639.       7.45       85.7  5.70e-267
#> 2 ratio         -0.649    0.354      -1.83 6.79e-  2
#> 3 income         1.84     0.0928     19.8  4.38e- 62
```

Testing for heteroskedasticity

Examples

Now, let's see what the residuals suggest about heteroskedasticity





Testing for heteroskedasticity

Example: Goldfeld-Quandt

Income looks potentially heteroskedastic; let's test via Goldfeld-Quandt.

Testing for heteroskedasticity

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Testing for heteroskedasticity

Example: Goldfeld-Quandt

Income looks potentially heteroskedastic; let's test via Goldfeld-Quandt.

```
#> [1] 19305.01
```

```
#> [1] 29537.83
```

Testing for heteroskedasticity

Example: Goldfeld-Quandt

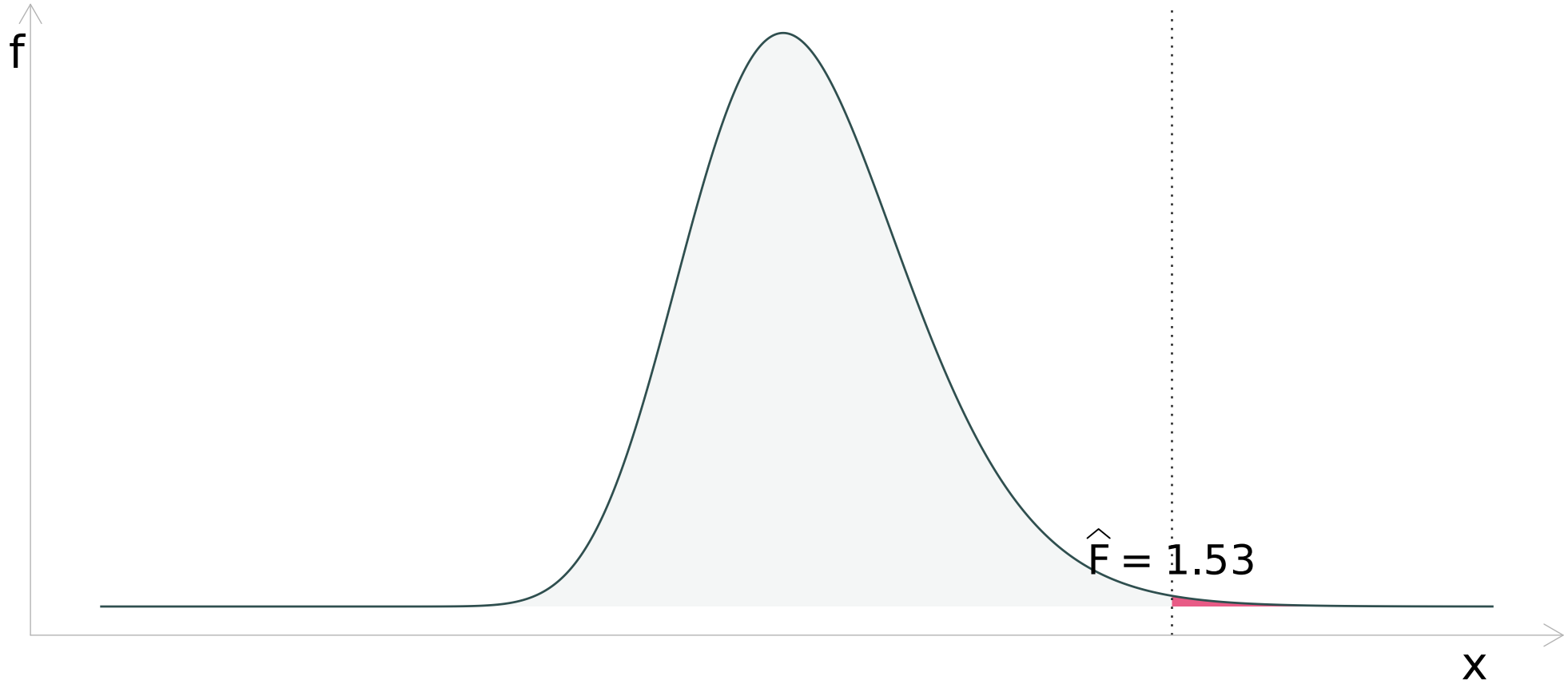
Remember the Goldfeld-Quandt test statistic?

$$F_{n^\star - k,\, n^\star - k} = \frac{\text{SSE}_2}{\text{SSE}_1}$$

Testing for heteroskedasticity

Example: Goldfeld-Quandt

The Goldfeld-Quandt test statistic and its null distribution



Testing for heteroskedasticity

Example: Goldfeld-Quandt

Putting it all together:

H_0: $\sigma_1^2 = \sigma_2^2$ *vs.* H_A: $\sigma_1^2 \neq \sigma_2^2$

Goldfeld-Quandt test statistic: $F \approx 1.53$

p-value ≈ 0.00423

$\therefore$ Reject H_0 (*p*-value is less than 0.05).

Conclusion: There is statistically significant evidence that $\sigma_1^2 \neq \sigma_2^2$. Therefore, we find statistically significant evidence of heteroskedasticity (at the 5-percent level).

Testing for heteroskedasticity

Example: Goldfeld-Quandt

What if we had chosen to focus on student-to-teacher ratio?

Testing for heteroskedasticity

Example: Goldfeld-Quandt

$$F_{n^\star - k,\, n^\star - k} = \frac{\text{SSE}_4}{\text{SSE}_3} \approx \frac{29,101.52}{26,243.52} \approx 1.11$$

which has a p-value of approximately 0.2603.

Heteroskedasticity

Example: White

Let's test the same model and data with the White test.

Recall: We saved our residuals as `e` in our dataset, *i.e.*,

Heteroskedasticity

Example: White

The **White test** adds squared terms and interactions to initial regression specification (the right-hand side)

$$
\begin{aligned}
u_i^2 = & \alpha_0 + \alpha_1 \text{Ratio}_i + \alpha_2 \text{Income}_i \\
& + \alpha_3 \text{Ratio}_i^2 + \alpha_4 \text{Income}_i^2 + \alpha_5 \text{Ratio}_i \times \text{Income}_i \\
& + w_i
\end{aligned}
$$

The **White test** tests the null hypothesis

H_0: $\alpha_1 = \alpha_2 = \alpha_3 = \alpha_4 = \alpha_5 = 0$

We just need to write some R code to test H_0.

Heteroskedasticity

Example: White

Aside: R has funky notation for squared terms and interactions in `lm()`:

- **Squared terms** use `I()`, *e.g.*, `lm(y ~ I(x^2))`
- **Interactions** use `:` between the variables, *e.g.*, `lm(y ~ x1:x2)`

Example: Regress `y` on quadratic of `x1` and `x2`:

Heteroskedasticity

Example: White

Step 1: Regress e_i^2 on 1st degree, 2nd degree, and interactions

Heteroskedasticity

Example: White

Step 2: Collect R_e^2 from the regression.

```
#> [1] 0.07332222
```

Heteroskedasticity

Example: White

Step 3: Calculate White test statistic $\mathbf{LM} = n \times R_e^2 \approx 420 \times 0.073$

```
#> [1] 30.79533
```

Heteroskedasticity

Example: White

Step 4: Calculate the associated *p*-value (where $\mathbf{LM} \overset{d}{\sim} \chi^2_k$); here, $k = 5$

```
#> [1] 1.028039e-05
```

Heteroskedasticity

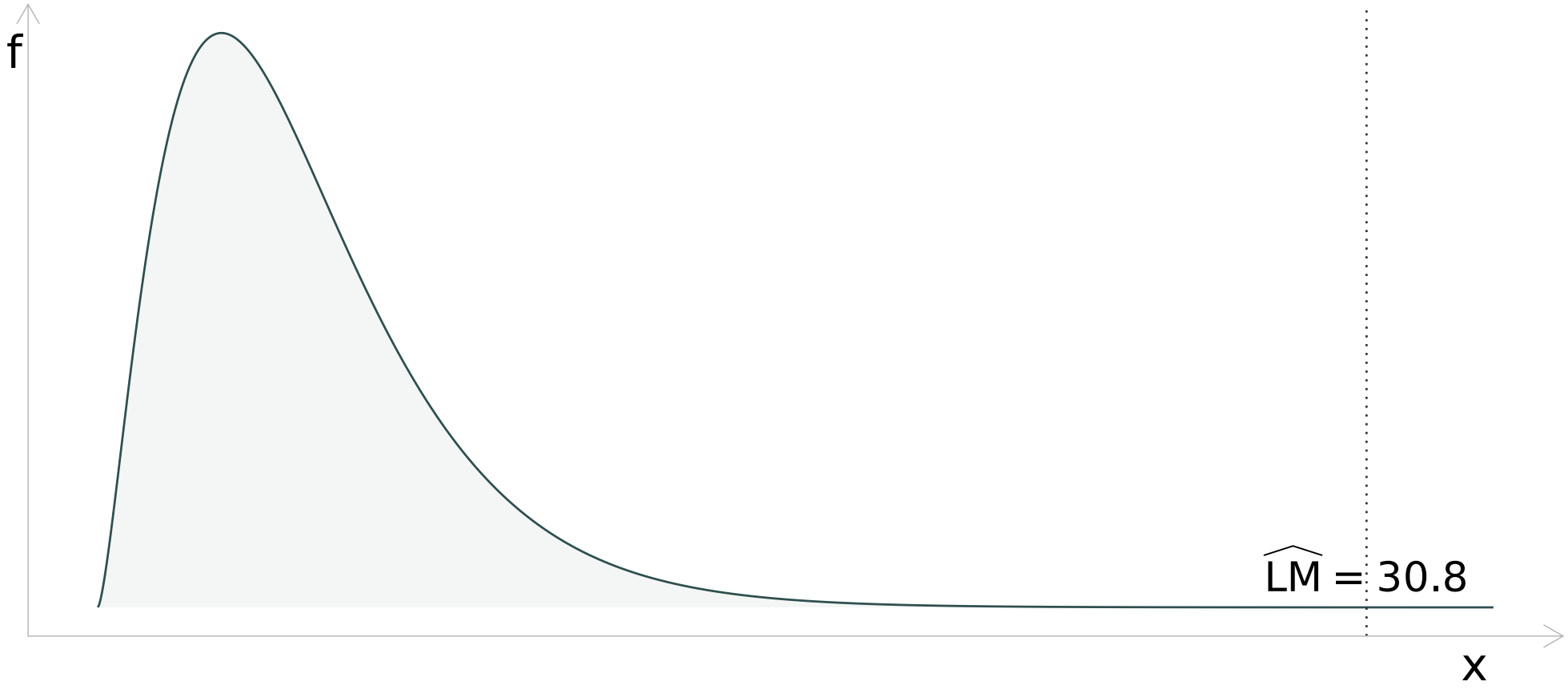
Example: White

Putting everything together...

Heteroskedasticity

Example: White

The White test statistic and its null distribution



Heteroskedasticity

Review questions

Heteroskedasticity

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- **Q:** What is the definition of heteroskedasticity?

Heteroskedasticity

Review questions

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- **Q:** Why are we concerned about heteroskedasticity?

Heteroskedasticity

Review questions

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- **Q:** Does plotting y against x, tell us anything about heteroskedasticity?

Heteroskedasticity

Review questions

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- **Q:** Does plotting e against x, tell us anything about heteroskedasticity?

Heteroskedasticity

Review questions

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- **Q:** Does plotting y against x, tell us anything about heteroskedasticity?
- **Q:** Does plotting e against x, tell us anything about heteroskedasticity?

- **Q:** Since we cannot observe the u_i's, what do we use to *learn about* heteroskedasticity?

Heteroskedasticity

Review questions

- **Q:** What is the definition of heteroskedasticity?
- **Q:** Why are we concerned about heteroskedasticity?
- **Q:** Does plotting y against x, tell us anything about heteroskedasticity?
- **Q:** Does plotting e against x, tell us anything about heteroskedasticity?
- **Q:** Since we cannot observe the u_i's, what do we use to *learn about* heteroskedasticity?

- **Q:** Which test do you recommend to test for heteroskedasticity? Why?

Next time: Living/working with heteroskedasticity.

Appendix

One more test...

Testing for heteroskedasticity

The Breusch-Pagan test

Breusch and Pagan (1981) attempted to solve this issue of being too specific with the functional form of the heteroskedasticity.

- Allows the data to show if/how the variance of u_i correlates with X.
- If σ_i^2 correlates with X, then we have heteroskedasticity.
- Regresses e_i^2 on $X = [1, x_1, x_2, \ldots, x_k]$ and tests for joint significance.

Testing for heteroskedasticity

The Breusch-Pagan test

How to implement:

1. Regress y on an intercept, x_1, x_2, ..., x_k.
2. Record residuals e.
3. Regress e^2 on an intercept, x_1, x_2, ..., x_k.

$$e_i^2 = \alpha_0 + \alpha_1 x_{1i} + \alpha_2 x_{2i} + \cdots + \alpha_k x_{ki} + v_i$$

4. Record R^2.
5. Test hypothesis H_0: $\alpha_1 = \alpha_2 = \cdots = \alpha_k = 0$

Testing for heteroskedasticity

The Breusch-Pagan test

The B-P test statistic[1] is

$$\mathrm{LM} = n \times R_e^2$$

where R_e^2 is the R^2 from the regression

$$e_i^2 = \alpha_0 + \alpha_1 x_{1i} + \alpha_2 x_{2i} + \cdots + \alpha_k x_{ki} + v_i$$

Under the null, LM is asymptotically distributed as χ_k^2.

1. This specific form of the test statistic actually comes form Koenker (1981).

Testing for heteroskedasticity

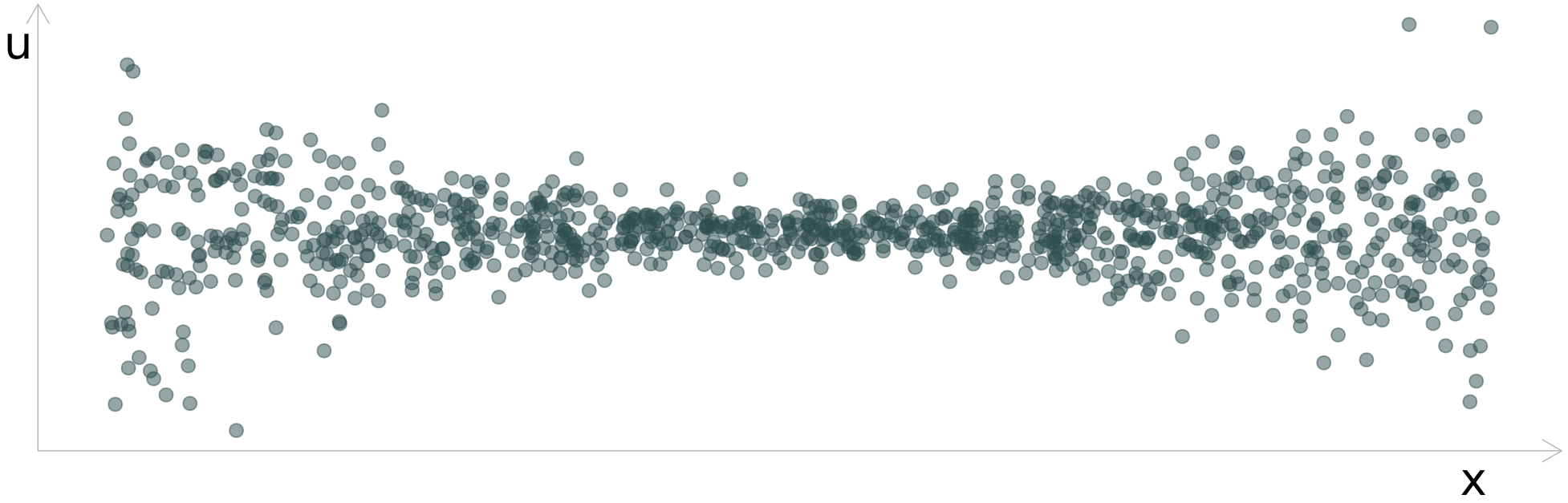
The Breusch-Pagan test

Problem: We're still assuming a fairly restrictive **functional form** between our explanatory variables $\boldsymbol{X}$ and the variances of our disturbances σ_i^2.

Testing for heteroskedasticity

The Breusch-Pagan test

Breusch-Pagan tests are still **sensitive to functional form**.



$$e_i^2 = \hat{\alpha}_0 + \hat{\alpha}_1 x_{1i} \qquad \widehat{\text{LM}} = 1.26 \qquad p\text{-value} \approx 0.261$$

$$e_i^2 = \hat{\alpha}_0 + \hat{\alpha}_1 x_{1i} + \hat{\alpha}_2 x_{1i}^2 \qquad \widehat{\text{LM}} = 185.8 \qquad p\text{-value} < 0.001$$

Testing for heteroskedasticity

Example: Breusch-Pagan

Let's test the same model with the Breusch Pagan.

Recall: We saved our residuals as `e` in our dataset, *i.e.*,

Testing for heteroskedasticity

Example: Breusch-Pagan

In B-P, we first regress e_i^2 on the explanatory variables,

Testing for heteroskedasticity

Example: Breusch-Pagan

and use the resulting R^2 to calculate a test statistic.

```
#> [1] 3.23205e-05
```

Testing for heteroskedasticity

Example: Breusch-Pagan

The Breusch-Pagan test statistic is

$$\mathrm{LM} = n \times R_e^2$$

1. k is the number of explanatory variables (excluding the intercept).

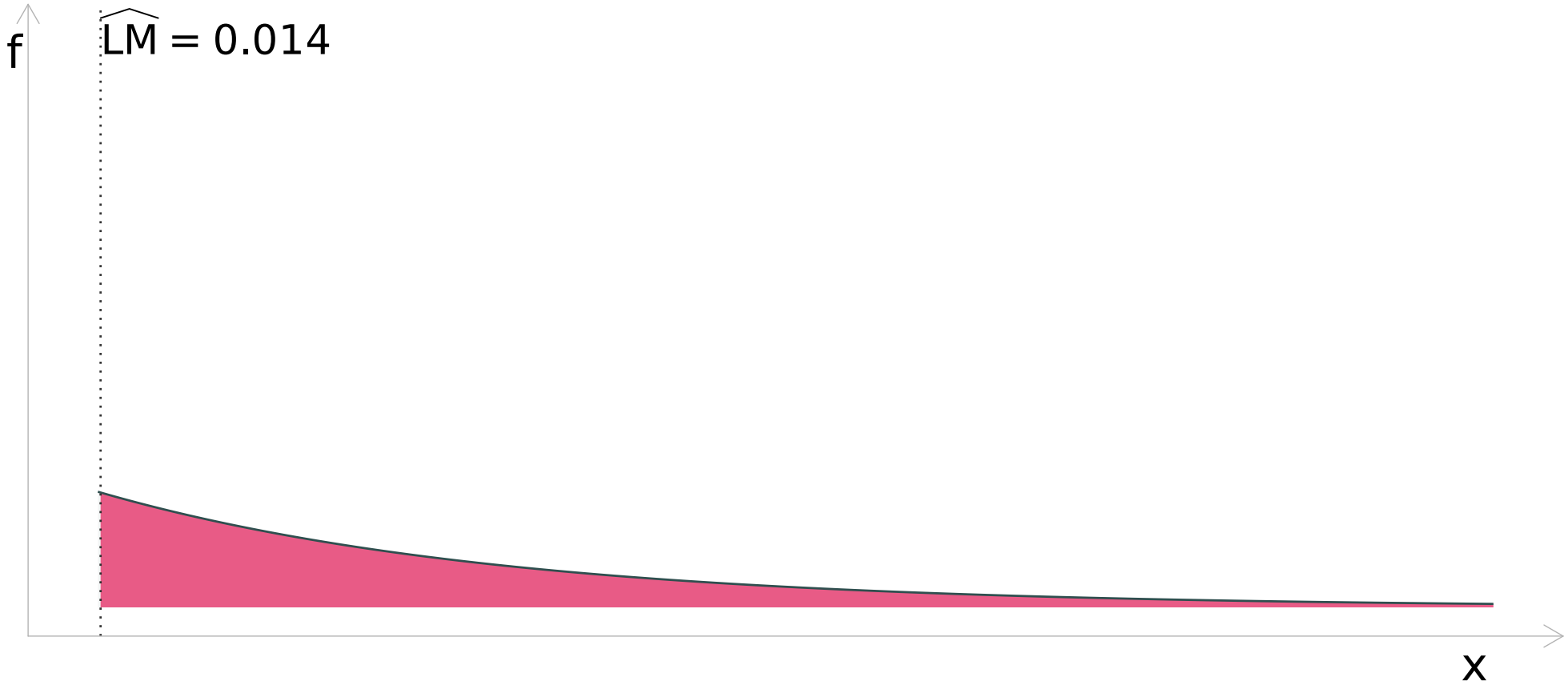
Testing for heteroskedasticity

Example: Breusch-Pagan

H_0: $\alpha_1 = \alpha_2 = 0$ *vs.* H_A: $\alpha_1 \neq 0$ and/or $\alpha_2 \neq 0$

for the model $u_i^2 = \alpha_0 + \alpha_1 \text{Ratio}_i + \alpha_2 \text{Income}_i + w_i$

Testing for heteroskedasticity

Example: Breusch-Pagan

The Breusch-Pagan test statistic and its null distribution